

SANKUR KUNDU

Software Engineer | Backend, Systems & Full-Stack

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SUMMARY

Software engineer (B.Tech CSE, 2026) who builds systems from scratch in Rust and proves them correct: **kiln**, an ML compiler faster than torch.compile on CPU; **ferrolm**, an LLM server; **quorumbd**, a distributed SQL database verified by simulation and TLA+; **faultline**, a Jepsen-style tester for etcd. Co-founded Gigzen and shipped alone its **three-app delivery platform**.

PROJECTS

kiln — Rust, zero dependencies: an ML compiler, ONNX to SIMD kernels

- Built from scratch an **ML compiler** (XLA/TVM-style): **ONNX** import, constant folding, **operator fusion**, an affine **loop IR**, **SIMD code generation**, an **auto-tuner** and a memory planner; output matches **PyTorch** on BERT and Llama.
- On 4 CPU cores, **faster than torch.compile**, **ONNX Runtime** and **JAX/XLA**: BERT **126 ms** vs 133, 146, 170; SmolLM2 **102 ms** vs 109, 111, 126; auto-tuning gains up to **1.31×**, fusion up to **1.20×**; **381 MB** of intermediates fit in a **16 MB** arena.
- CI checks its machine code against **PyTorch** with each optimisation switched off and under random schedules.

ferrolm — Rust, zero dependencies: an LLM inference server and RAG stack

- Built from scratch an **OpenAI-compatible LLM server** with a **paged KV cache**, **continuous batching**, **prefix caching** and **speculative decoding** on hand-written **SIMD kernels**; runs SmolLM2 and Qwen2.5, matching **Hugging Face transformers**.
- On 4 CPU cores: **3.4×** throughput (continuous batching), first token **46×** sooner (prefix caching), **20×** more requests in the same memory (paging), **1.47×** faster decoding at identical output (speculation).
- Extended it into a **RAG stack**: **int8/int4 quantization** with **AWQ** (**1.86×** faster), **BERT embeddings** matching transformers, **HNSW / IVF-PQ** search at **FAISS**-level recall on 1M vectors; retrieval lifts SQuAD F1 **15 → 48**.

quorumbd — Rust, zero dependencies: a distributed SQL database, PostgreSQL wire-compatible

- Built from scratch a **distributed SQL database** that **psql** connects to: an **LSM storage engine**, **Raft consensus**, a **sharded multi-Raft store** that splits and rebalances itself, **serializable ACID transactions** (Percolator) and a SQL planner; **1,500 tpmC** on a **TPC-C**-derived benchmark, consistency checks passing.
- Proved by **deterministic simulation** (**10,000 Raft clusters**, **610K crashes**, histories machine-checked) and a **TLA+** model (**337K states**); found **6 real bugs**, catches all **15 planted**; **time-travel debugger** rewinds runs by replay.

faultline — Rust, zero dependencies: Jepsen-style fault injection for etcd

- Runs real **etcd** clusters in **Linux network namespaces**, injects **iptables partitions**, **kill -9** crashes and **SIGSTOP** pauses, and checks histories for **linearizability**, **watch completeness** and **lost updates**.
- Tested **4 etcd release lines** with **1.26M operations** under **288 faults**: linearizability and watches held; traced **lost updates** under etcd locks to a paused leader applying a timed-out write **6 s** late, and showed **fenced writes** prevent it.

EXPERIENCE

Gigzen — Co-founder & Founding Engineer

Apr 2026 – Present

Bhubaneswar, India · Early-stage startup (incorporation in progress) · Sole engineer across every product

- Sole engineer** on **Waggle**, a delivery platform of **three Android apps** (riders, shops, customers) from one **TypeScript / React / Capacitor** codebase on **Supabase PostgreSQL**: **53 tables**, **90 RLS policies**, **154 SQL functions**.
- Put the business rules in the database, where no client can bypass them: **first-accept-wins dispatch** (45-second offer window), **4-digit handover codes**, **Razorpay/UPI** payments reconciled by UTR, **GST invoices** and **pg_cron** billing runs.
- Security-tested** the platform as one system: **150 automated attacks** on the database access rules, all blocked; a **42-step Playwright** end-to-end run across all three apps, from rider verification to delivery rating.
- Built **Populace**, an open-source multi-user testing tool (**Node.js**, **zero runtime dependencies**) shipped as an **npm CLI**, a **GitHub Action** and an **Electron** desktop app. Its **first run found 5 defects in 3.5 minutes** after full manual testing had passed, including one that silently broke **every new signup**.
- Designed its **13-method adapter contract** and proved it on third-party software (**Gitea**: 1,106 API calls); **3.17M API calls** across 30–300 simulated users with **zero API failures**; **122 self-tests** in CI.

SKILLS

CS Fundamentals: Data Structures & Algorithms, System Design, Distributed Systems, DBMS internals, OS, Networks

Languages: Rust, TypeScript, JavaScript, Python, SQL (PL/pgSQL), Java

Backend & Data: PostgreSQL, Supabase, Row-Level Security, pg_cron, REST APIs, Node.js, FastAPI, Flask, Redis, Razorpay/UPI

Frontend & Mobile: React, Capacitor (Android), Electron, Leaflet/MapLibre maps, HTML, CSS

Systems & performance: ML compilers (fusion, code generation, auto-tuning), SIMD (AVX-512), LLM serving, profiling

Testing, DevOps & Cloud: Simulation testing, fault injection, TLA+, Playwright, GitHub Actions, Docker, Kubernetes, GCP

EDUCATION

B.Tech, Computer Science and Engineering (AI & ML) — IEM Kolkata

Sep 2022 – Apr 2026